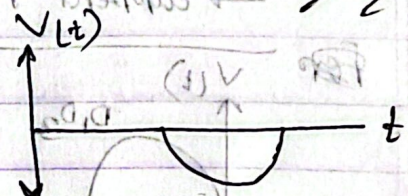
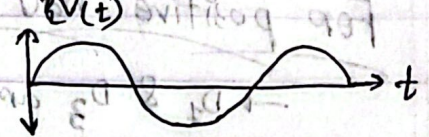
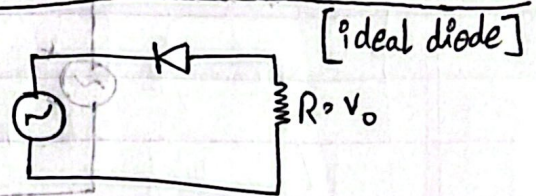
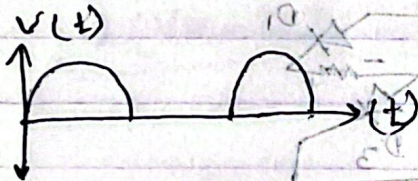
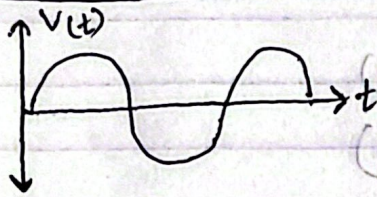
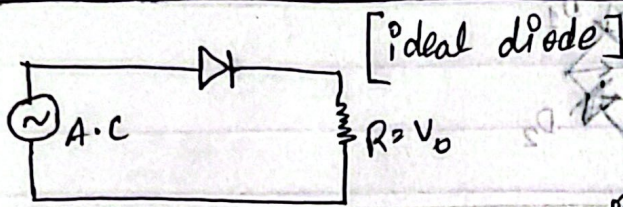


Half wave rectifier (Considering an ideal diode)

Converts A.C to pulsating D.C by allowing only one-half cycle of the A.C waveform to pass.



$$V_{dc} / V_{avg} = \frac{1}{2\pi} \left[\int_0^{\pi} V_m \sin(\theta) d\theta + \int_{\pi}^{2\pi} 0 d\theta \right]$$

$$= \frac{1}{2\pi} \int_0^{\pi} V_m \sin(\theta) d\theta$$

$$= \frac{V_m}{2\pi} [-\cos(\theta)]_0^{\pi}$$

$$= \frac{V_m}{2\pi} [-\cos(\pi) + \cos(0)]$$

$$= \frac{V_m}{2\pi} [1 + 1]$$

$$= \frac{2V_m}{2\pi}$$

$$= \frac{V_m}{\pi}$$

$$\frac{V_{dc}}{V_{avg}} = \frac{1}{2\pi} \left[\int_0^{\pi} 0 d\theta + \int_{\pi}^{2\pi} V_m \sin(\theta) d\theta \right]$$

$$= \frac{1}{2\pi} \int_{\pi}^{2\pi} V_m \sin(\theta) d\theta$$

$$= \frac{V_m}{2\pi} [-\cos(\theta)]_{\pi}^{2\pi}$$

$$= \frac{V_m}{2\pi} [-\cos(2\pi) + \cos(\pi)]$$

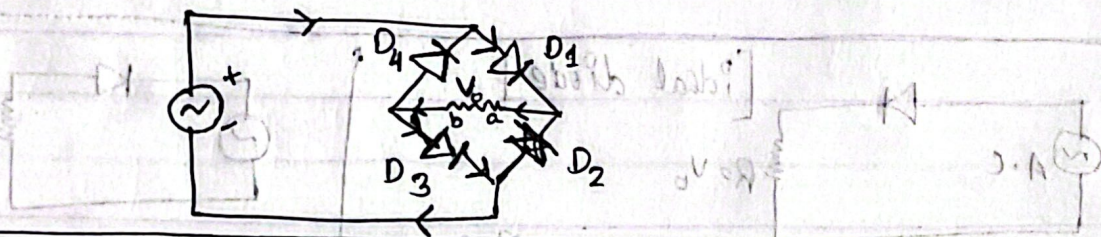
$$= \frac{V_m}{2\pi} [-1 - 1]$$

$$= -\frac{2V_m}{2\pi}$$

$$= -\frac{V_m}{\pi}$$

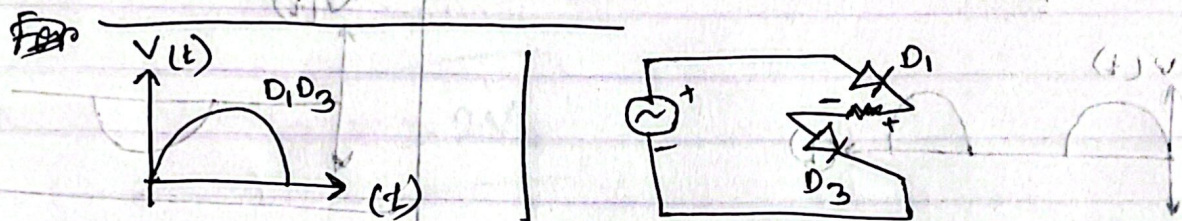
Bridge Rectifier

Turns A.C. to pulsating D.C. for both half cycle



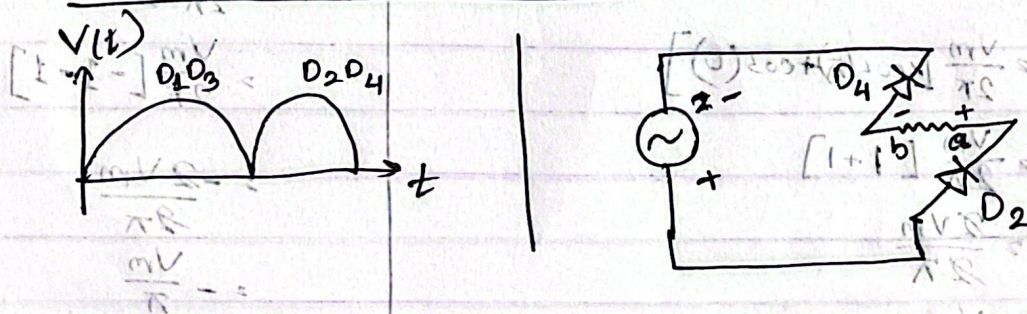
For positive half cycle:

- D_1 & D_3 are forward bias (short)
- D_2 & D_4 are reversed bias (open)
- current flow, $a \rightarrow b$



For negative half cycle:

- D_2 & D_4 are forward bias (short)
- D_1 & D_3 are reversed bias (open)
- current flow, $a \rightarrow b$



PIV: So diode never goes negative.

↳: Giving a diode ~~higher~~ voltage ~~than~~ higher than the breakdown voltage, during manufacturing using doping process.

For negative half cycle:-

$$V_m = \frac{1}{2\pi} \left[\int_0^{\pi} V_m \sin(\theta) d\theta + \int_{\pi}^{2\pi} V_m \sin(\theta) d\theta \right]$$

$$= 2 \times \frac{1}{2\pi} \int_0^{\pi} V_m \sin(\theta) d\theta$$

$$= \frac{V_m}{\pi} [-\cos(\theta)]_0^{\pi}$$

$$= \frac{V_m}{\pi} [-\cos(\pi) - (-\cos(0))]]$$

$$= \frac{V_m}{\pi} [-(-1) - (-1)]$$

$$= \frac{2V_m}{\pi} \text{ [Ideal diode]}$$

$$= \frac{2(V_m - 0.7 - 0.7)}{\pi} \text{ [Practical silicon diode]}$$

$$= \frac{2(V_m - 1.4)}{\pi} \text{ [Practical silicon diode]}$$

For positive half cycle:

$$V_m = \frac{1}{2} \left[\int_0^{\pi} V_m \sin(\theta) d\theta + \int_{\pi}^{2\pi} V_m \sin(\theta) d\theta \right]$$

$$= 2 \times \frac{1}{2\pi} \int_{\pi}^{2\pi} V_m \sin(\theta) d\theta$$

$$= \frac{V_m}{\pi} [-\cos(\theta)]_{\pi}^{2\pi}$$

$$= \frac{V_m}{\pi} [-(-1) - (-(-1))]]$$

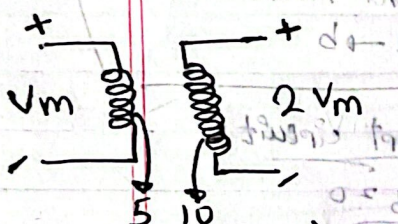
$$= \frac{V_m}{\pi} [-1 - 1]$$

$$= -\frac{2V_m}{\pi} \text{ [ideal diode]}$$

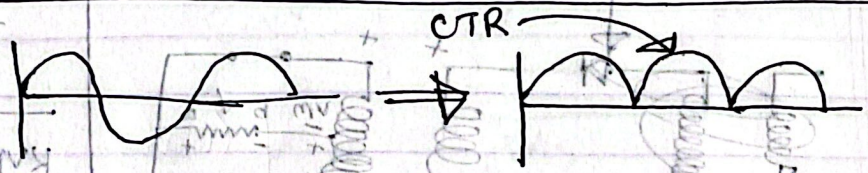
$$= -\frac{2(V_m - 1.4)}{\pi} \text{ [Practical silicon diode]}$$

Centered tapped rectifier (2 diode)

A CTR is a circuit that converts AC into pulsating DC using a transformer with a center-tapped secondary winding and two diodes to convert both halves of AC waveform.



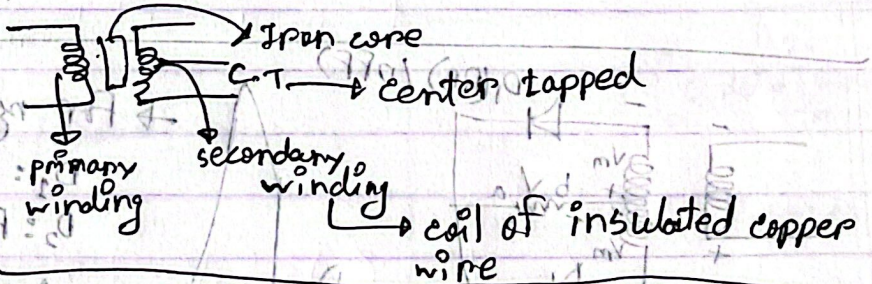
1:2 (Fixed)
(polarity of wires have to be same)



Transformer is a electric device that transfers AC electrical energy from one circuit to another.

Used to step up or step down AC voltage while keeping frequency constant.

pulsating DC
purely positive



Iron core concentrates and guides the magnetic flux to the secondary, instead of letting the magnetic field spread through air.

Electric current → magnetic field

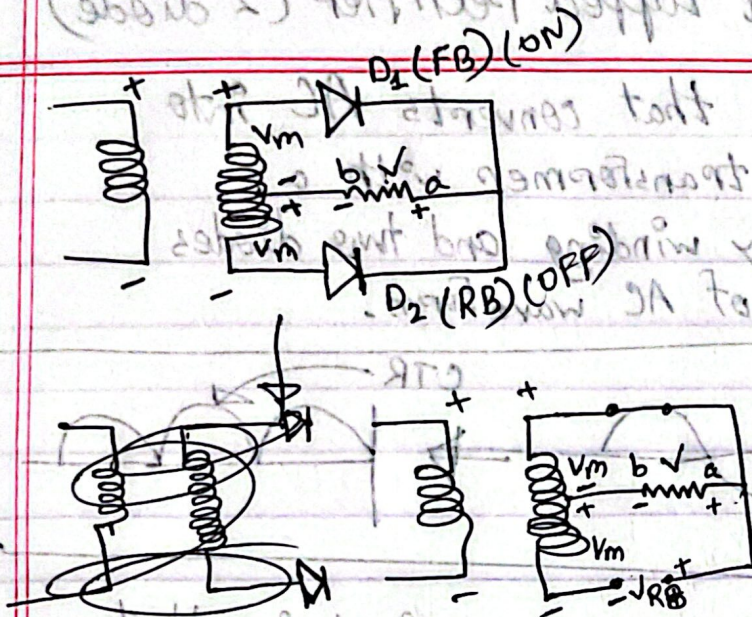
voltage	Electrical push	Volt
current	Flow of electric charge	Ampere
Magnetic flux	Amount of magnetic field through an area	Weber

$$1 \text{ kV} = 1000 \text{ V}$$

1 kV is 1000 V

$$1 \text{ V} = 1000 \text{ mV}$$

(circuit 2) rectifier bridge



→ For positive half cycle:

D_1 : forward bias

D_2 : Reverse bias

→ Current Flow:

$a \rightarrow b$

KVL in short circuit:

$$-V_m + V_{ab} = 0$$

$$\Rightarrow V_{ab} = V_m$$

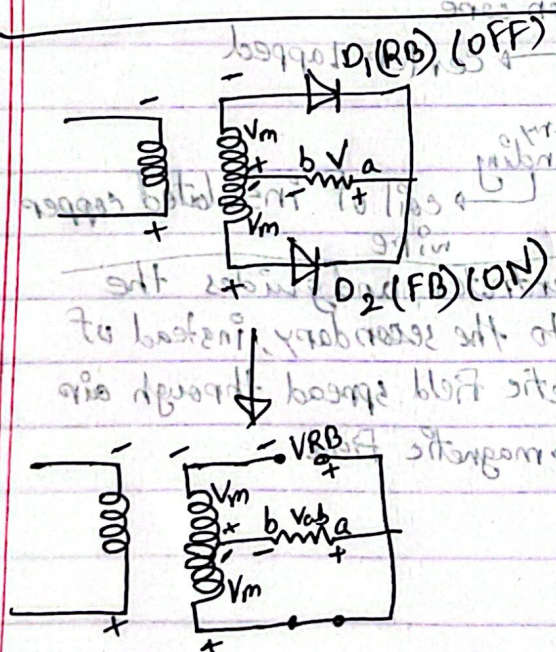
KVL in open circuit:

$$-V_m - V_{ab} + V_{RB} = 0$$

$$\Rightarrow -V_m - V_m + V_{RB} = 0$$

$$\Rightarrow V_{RB} = 2V_m$$

$$PIV > 2V_m$$



→ For negative half cycle

D_1 : Reverse Bias

D_2 : Forward Bias

→ Current Flow:

$a \rightarrow b$

KVL in short circuit:

$$+V_m - V_{ab} = 0$$

$$\Rightarrow V_m = V_{ab}$$

KVL in open circuit

$$+V_m - V_{RB} + V_{ab} = 0$$

$$\Rightarrow V_m - V_{RB} + V_m = 0$$

$$\Rightarrow V_{RB} = 2V_m$$

$$PIV > 2V_m$$